RoboReviews: AI-Powered Amazon Review Intelligence for Product Recommendations

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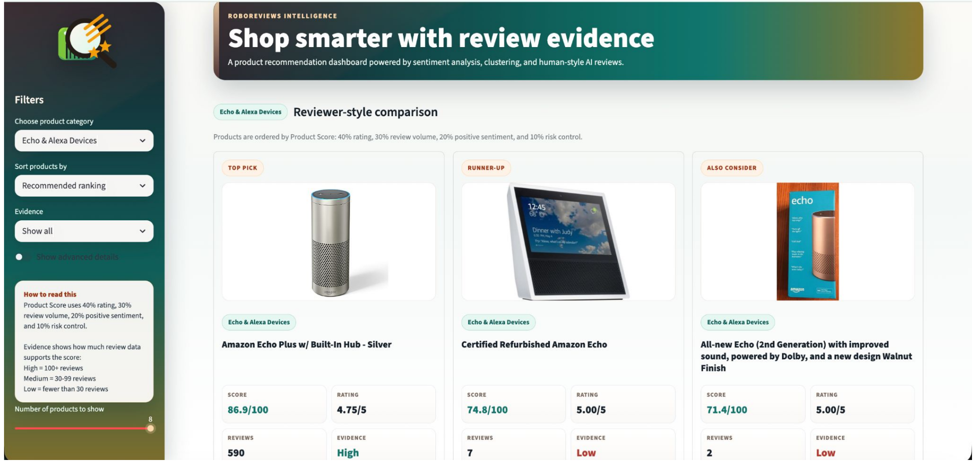
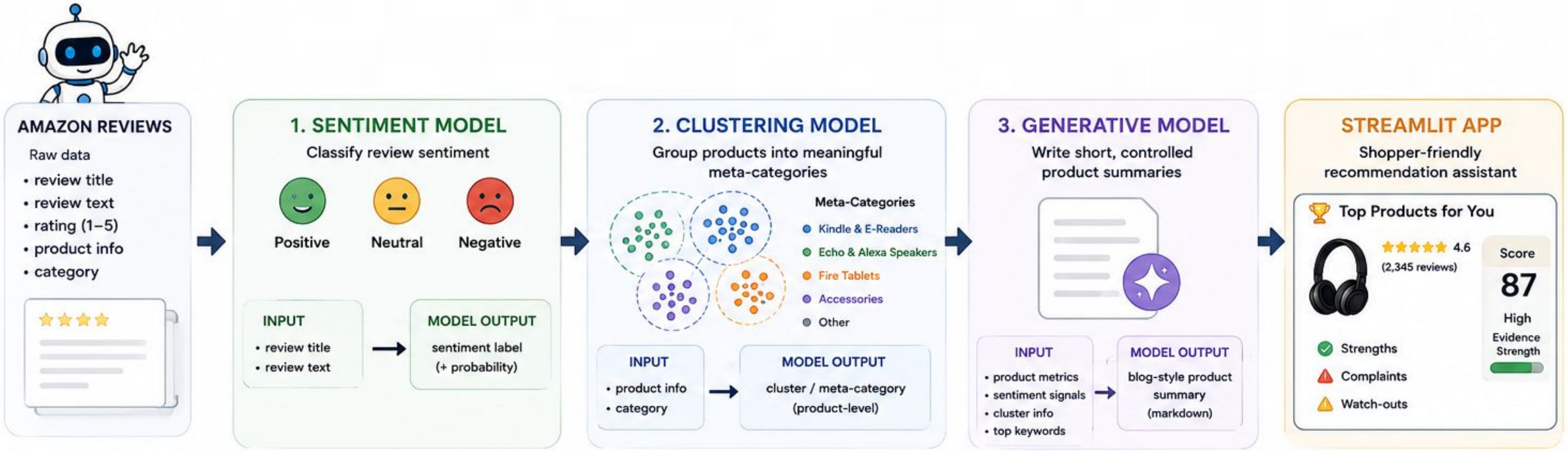
RoboReviews: Shop Smarter, Read Less

- Reads thousands of reviews
- Groups similar products
- Detects sentiment and complaints
- Ranks products by trust signals
- Generates efficient product summaries



RoboReviews: Pipeline Workflow



1 SENTIMENT MODEL

Goal: Predict the sentiment of each review
(Positive / Neutral / Negative)

METHODS

Input
Review title + review text
(rating used only as weak label)

Preprocessing
- Combine title + text
- Clean text (lowercase, remove punctuation, stopwords)
- Stemming (best)
- Tested: lemmatization

Vectorization
- TF-IDF with n-grams (1–2)
- Max features: 50,000
- Sublinear TF: True
- Min df: 1

Models Compared
Logistic Regression, **Linear SVM**
Naive Bayes, Random Forest, XGBoost

Tuning & Evaluation
- Stratified train/test split
- 3-fold cross-validation (train set)
- RandomizedSearchCV (TF-IDF + Linear SVM)
- Metric: Macro F1 (main)

BEST TUNED MODEL

Best preprocessing: stemming
Best model: **Linear SVM**

Best parameters:
- tfidf__sublinear_tf: True
- tfidf__ngram_range: (1, 2)
- tiuff__min_df: 1
- tiidf__max_features: 50000
- svm__class_weight: balanced
- svm__C: 1.0

TEST PERFORMANCE (Optimal Model)

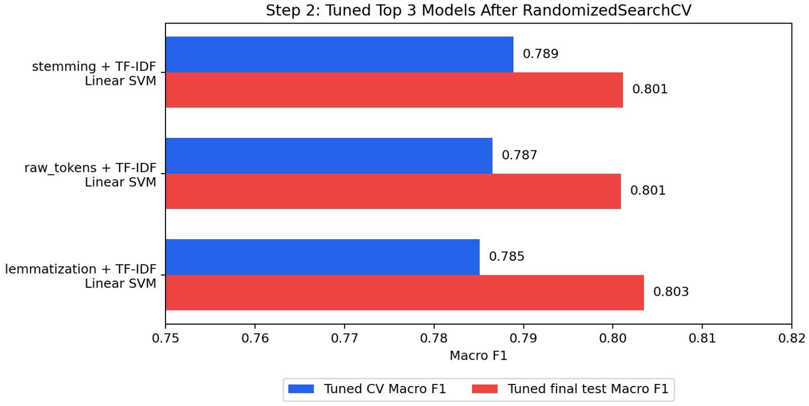
Metric	Score
CV Macro F1	0.789
Accuracy	0.963
Macro F1	0.801
Weighted F1	0.961

CLASS-LEVEL RESULTS (Test Set)

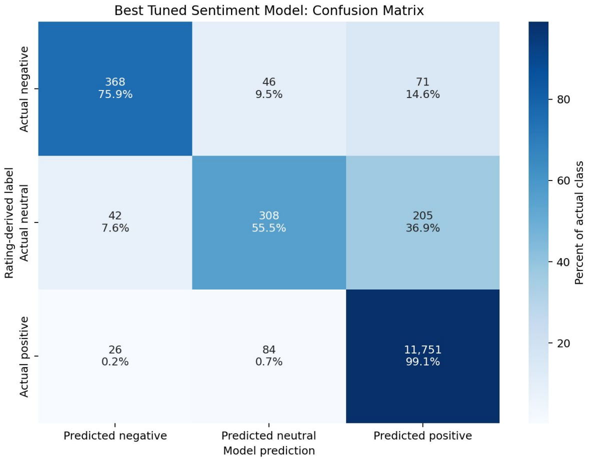
Class	Precision	Recall	F1-score	Support
Negative	0.84	0.76	0.80	485
Neutral	0.70	0.55	0.62	555
Positive	0.98	0.99	0.98	11,861
Accuracy	–	–	0.963	12,901
Macro avg	0.84	0.77	0.80	12,901
Weighted avg	0.96	0.96	0.96	12,901

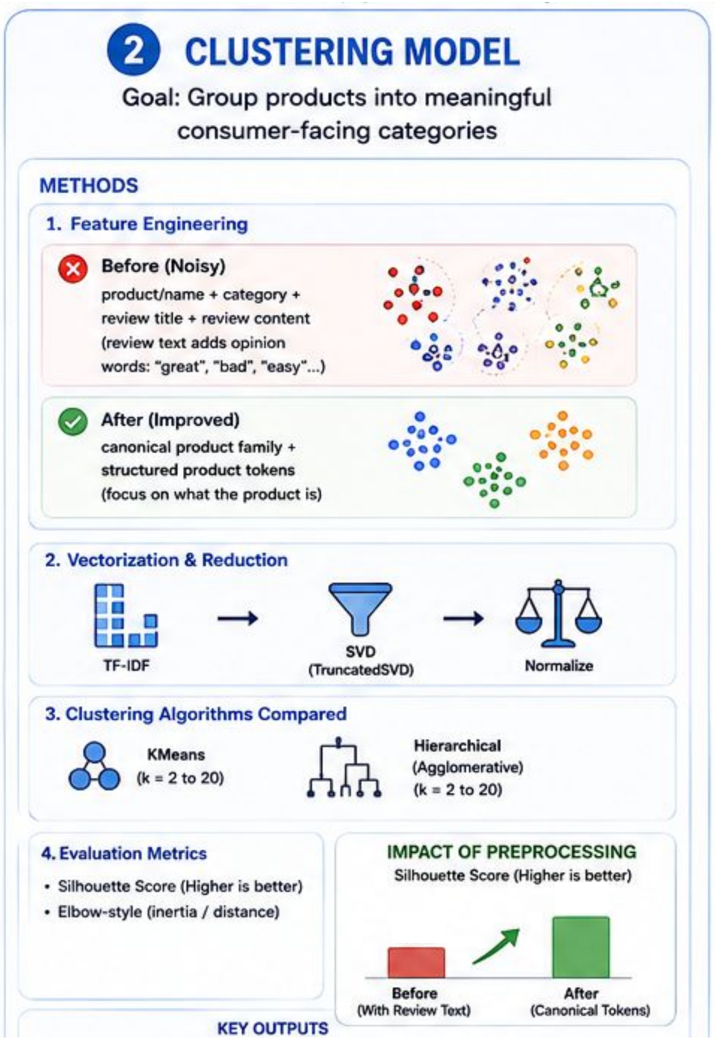
KEY INSIGHTS

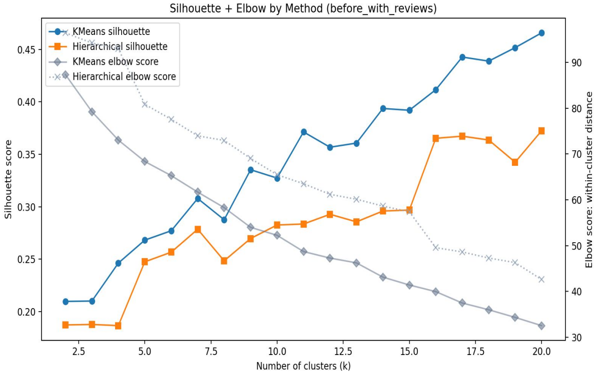
- Macro F1 matters more than accuracy because positive reviews dominate.
- Neutral class is hardest due to mixed opinions in 3-star reviews.

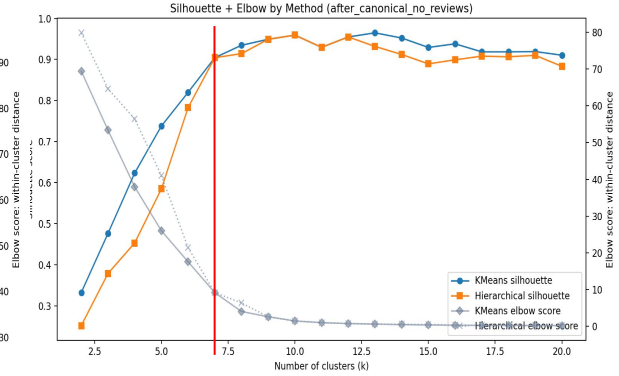


Both bars are after tuning: blue = tuned CV selection score, red = tuned final test score.

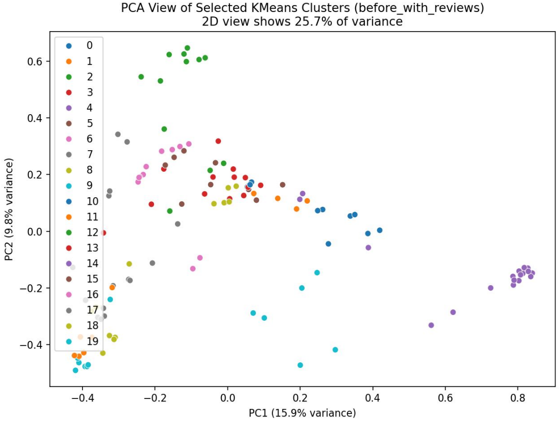


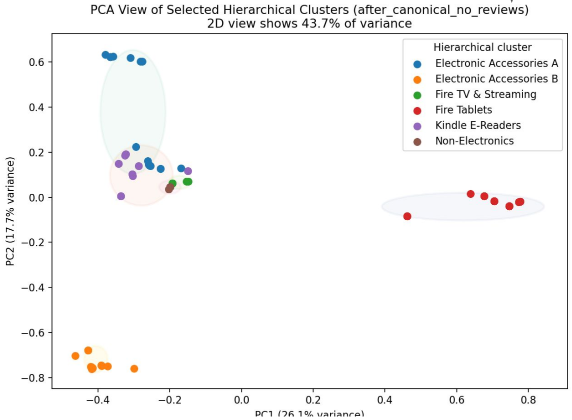






- **Selected k=6:** cleaner, easier-to-explain product groups.
- **k=7 scored higher**, but mainly over-split one category.
- **k=6 was more useful** for the recommendation app.





3 GENERATIVE MODEL (QWEN)

Turn structured review evidence into short, reviewer-style recommendation articles

Qwen

QWEN PARAMETERS

local Ollama model qwen2.5:

Temperature:	0.65	• Balanced creativity and control. • Not too robotic, but not too random.
Top-p:	0.90	• Allows natural wording while keeping output focused.
Repeat penalty:	1.12	• Reduces repeated phrases.
Max generated tokens (num_predict):	580	• Enough for a 3-paragraph mini review plus pros/cons.
Timeout:	180 seconds	• Allows slower local generation to finish.

PROMPT STRUCTURE (EVIDENCE GIVEN TO QWEN)

We gave Qwen **structured evidence**, not raw reviews.

- Product name
- Corrected category
- Product type
- Review count
- Average rating
- Positive review share (%)
- Negative review share (%)
- Evidence level (High / Medium / Low)
- Main strengths
- Main concerns
- Comparison notes inside the category

PROMPT RULES (FOR QWEN)

- Output must follow these fixed labels:
  HEADLINE | SUMMARY | BEST_FOR | PROS | CONS
- SUMMARY must be exactly **3 paragraphs.**
- Each paragraph: **~60–75 words.**
- Total summary length: **~190–220 words.**
- Mention the exact review count, rating, and positive share **ONLY ONCE**.
- Do NOT invent facts, prices, warranty claims, or outside competitors.
- Write helpful, human reviewer-style language (not data-report style).
- Include **product comparison insights:** explain how this product compares to other top products in the same category (strengths, weaknesses, best alternative when relevant).

{ } PROMPT EXAMPLE (JSON STRUCTURE)


```
{
  "product_name": "Echo Dot (5th Gen)",
  "corrected_category": "Smart Speakers",
  "product_type": "Smart Speaker",
  "review_count": 2345,
  "average_rating": 4.6,
  "positive_review_share": 0.89,
  "negative_review_share": 0.07,
  "evidence_level": "High",
  "main_strengths": ["Sound quality", "Easy setup", "Alexa integration", "Value for money"],
  "main_concerns": ["Occasional connectivity issues", "Bass could be stronger"],
  "comparison_notes": "Strong upgrade over Echo Dot (4th Gen) with better sound
      and improved sensors. Good value for the price.",
  "target_audience": "Users who want an affordable, easy-to-use smart speaker
      for home and daily tasks."

}
```


This structured evidence is placed in the prompt for Qwen to write the article.

HOW WE IMPROVED QWEN OUTPUT

- Added product-type detection (tablet, e-reader, smart speaker, accessory, etc.).
- Filtered strengths & concerns based on product type.
- Special handling for batteries/accessories to avoid irrelevant talk (e.g., screens, apps).
- Added few-shot examples for different product types.
- Stricter wording rules to avoid "data report" or generic claims.
- Required an evidence sentence to force accurate numbers.
- Added product comparison instruction (see rules).

FINAL RESULT:

Qwen became a writing layer, not the decision-maker.

The model writes friendly reviewer-style summaries using evidence from sentiment analysis, clustering, ratings, review volume, and product ranking.

Takeaway

- Star ratings alone do not tell the full story, reviews are noisy, mixed, and sometimes contradictory.
- RoboReviews combines sentiment, clustering, evidence strength, and complaint risk to create better product signals.
- Sentiment worked best with n-grams, and clustering worked best with canonical product identity.
- Qwen did not guess, it wrote summaries from structured evidence instead of raw review chaos.
- The result is a clearer, more trustworthy way to choose products.

RoboReviews Demo

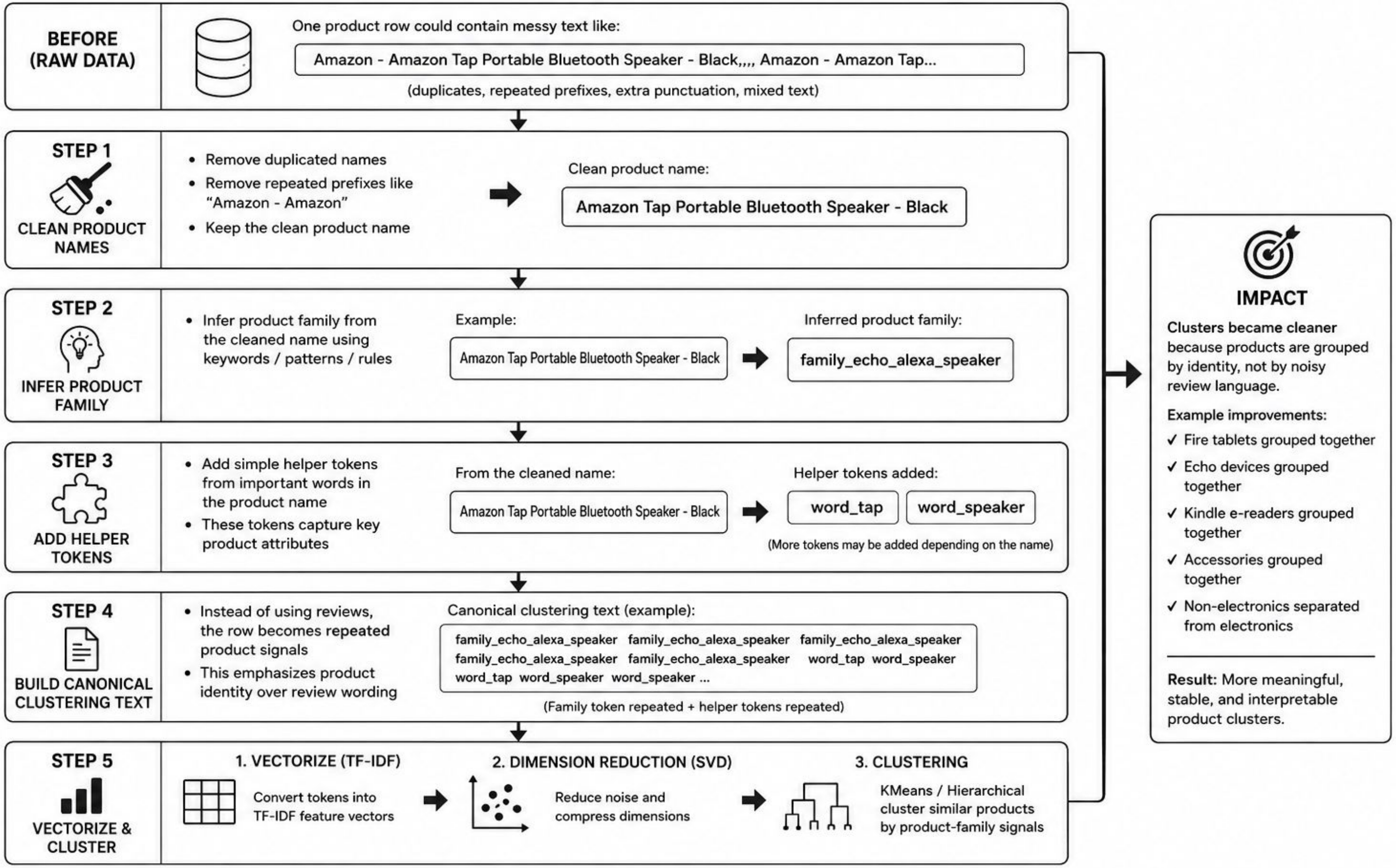
Thank you for the attention!

Recommandation Scores

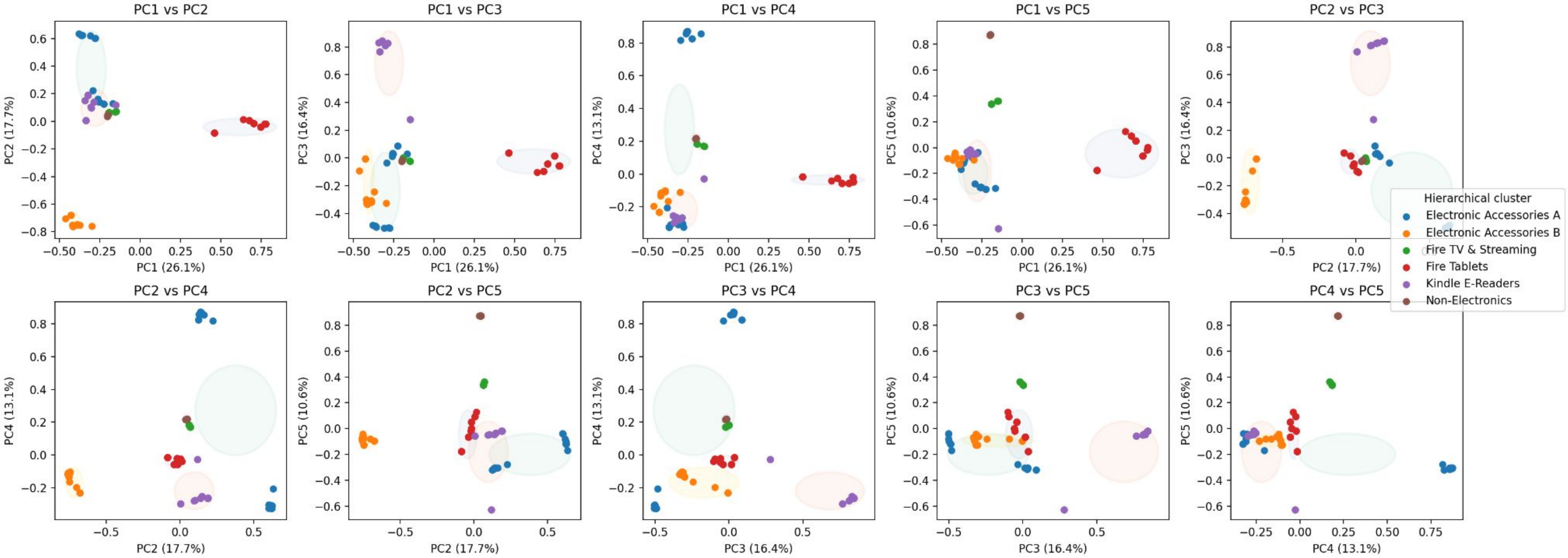
- **Product Score:** the app ranks products using a simple weighted score:
  - **40% average rating:** how customers scored the product.
  - **30% review volume:** more reviews give stronger evidence.
  - **20% positive sentiment:** how often review text sounds positive.
  - **10% risk control:**
    - Negative rate = number of negative reviews / total number of reviews
    - Risk control = 1 - negative rate
  - **Main idea:** the best product should have strong ratings, many reviews, positive review text, and fewer complaints.
- **Evidence strength:** the app checks how many reviews support each product:
  - **High evidence:** 100+ reviews.
  - **Medium evidence:** 30–99 reviews.
  - **Low evidence:** fewer than 30 reviews.
  - **Main idea:** products with more reviews are more reliable because the recommendation is based on more customer feedback.

CANONICAL CLUSTERING PREPROCESSING PIPELINE

Goal: make product clustering depend on product identity, not noisy review text.



PCA Cluster Views - After Canonical No Reviews



Each panel shows a different pair of PCA axes. Ellipses show 6 Hierarchical clusters; point colors and legend show readable cluster names. PCA is only for visualization.